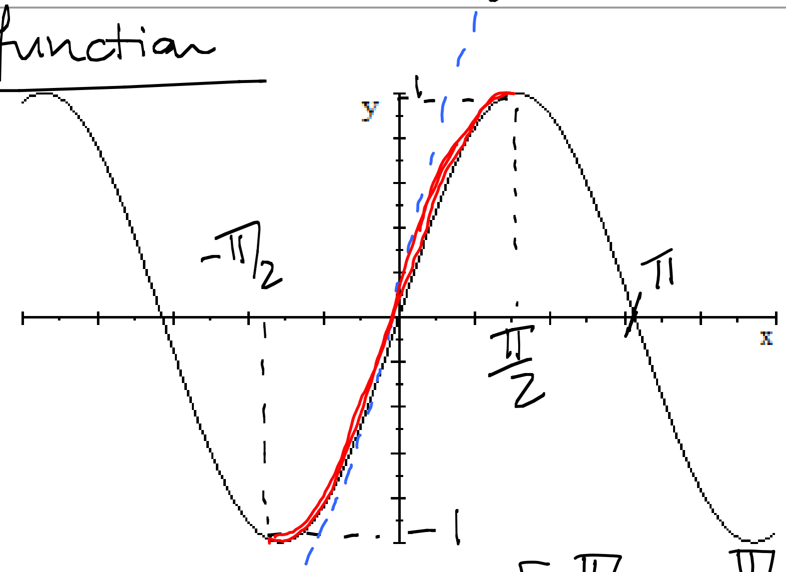
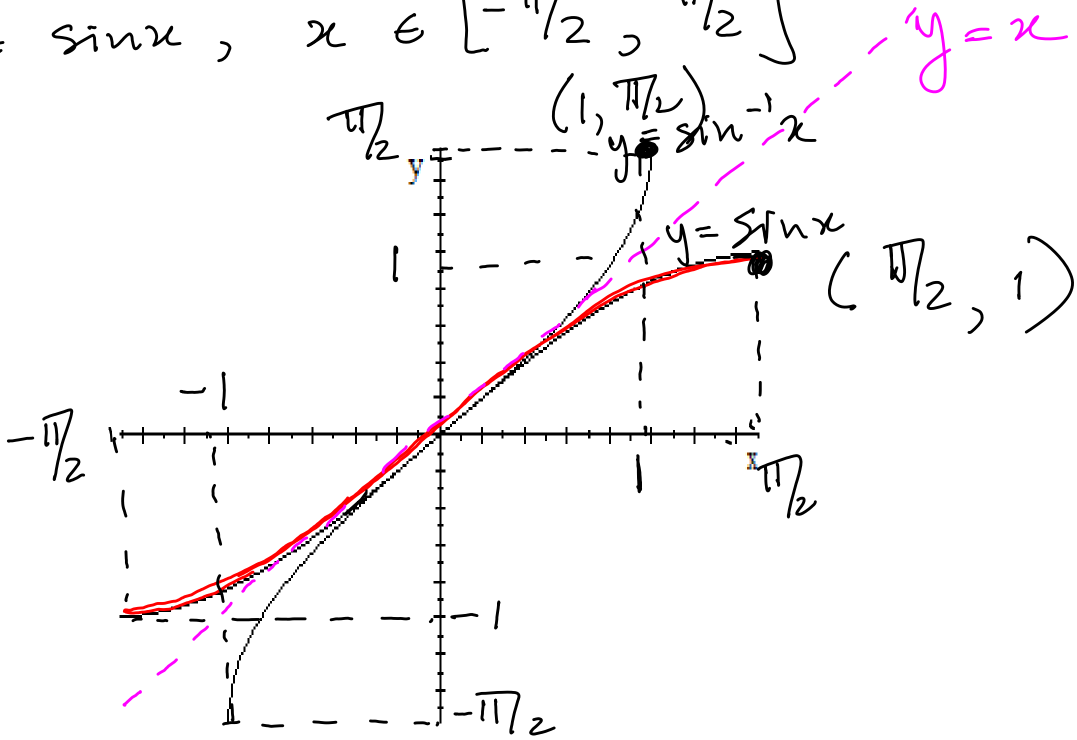
arcsin function

$y = \arcsin x$
$= \sin^{-1} x$



$y = \sin x, \quad x \in [-\pi/2, \pi/2]$



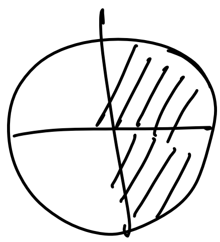
$y = \sin x, \quad x \in [-\pi/2, \pi/2]$

For the inverse :

$\underset{\text{value}}{x} = \sin \underset{\text{angle}}{y}, \quad y \in [-\pi/2, \pi/2]$

$\Rightarrow \underset{\text{angle}}{y} = \arcsin x, \quad y \in [-\pi/2, \pi/2]$

$= \sin^{-1} x$. value

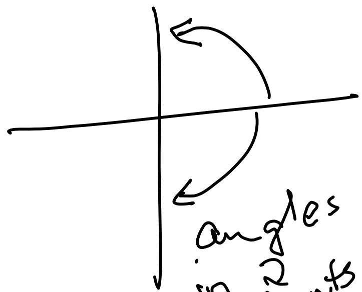


y is the angle for which the sine value is x.

Definition: $y = \arcsin x = \sin^{-1} x$

$$\iff \sin y = x \quad , \; y \in [-\pi/2, \pi/2]$$

$$y = \arcsin \tfrac{1}{2} \Rightarrow y = \pi/6$$



angles in 2 quadrants only

$$y = \sin^{-1}\left(-\frac{1}{\sqrt{2}}\right) = -\pi/4$$

$$\sin^{-1}(1) = \pi/2$$

$\arcsin(3)$ Does not exist

$\arcsin(\pi/2)$ Does not exist

value

angle

Cancellation properties:

$\sin(\arcsin x) = x$ for $x \in [-1, 1]$

$\arcsin(\sin x) = x$ for $x \in [-\pi/2, \pi/2]$

eg. $\sin(\arcsin \frac{1}{2}) = \frac{1}{2}$

Problems

1. $\sin(\arcsin(-1/2)) = \sin(-\pi/6) = -1/2$.

2. $\sin^{-1}(1) = \pi/2$

3. Solve $\sin x = \frac{1}{2} \Rightarrow x = \frac{\pi}{6} + 2k\pi$ or
$x = 5\pi/6 + 2k\pi$

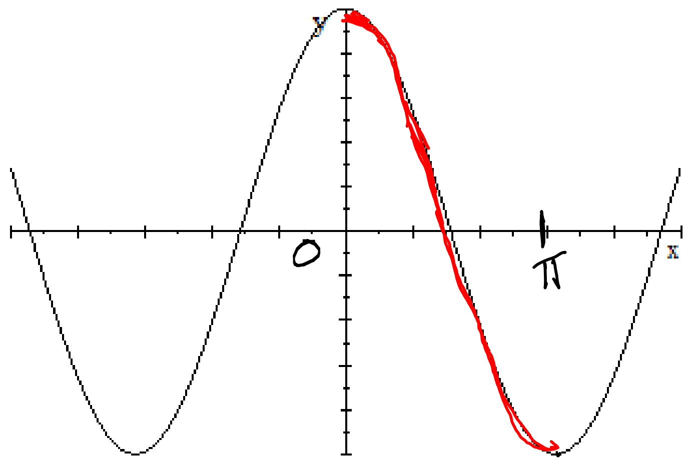
4. Write $y = \arcsin \frac{\sqrt{3}}{2}$ in terms of sin
$\Rightarrow \sin y = \sqrt{3}/2$

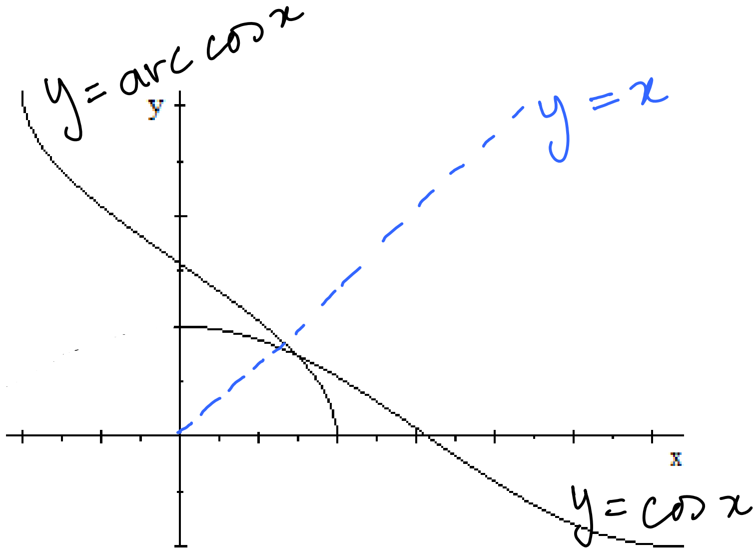
5. $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) = -\pi/3$

6. $\sin^{-1} 0 = 0$

7. $\arcsin\left(\frac{1}{\sqrt{2}}\right) = \pi/4$.

arccos x function





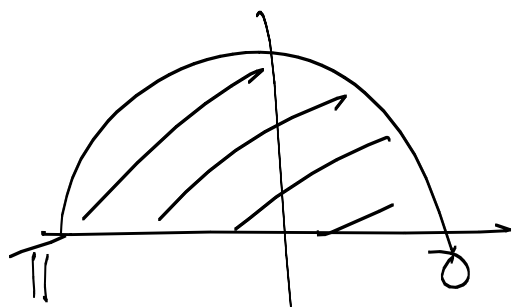
$$y = \cos x, \qquad x \in [0, \pi]$$

For the inverse:

$$x = \cos y, \qquad y \in [0, \pi]$$

$$\iff y = \arccos x, \qquad x \in [-1, 1]$$

$$\arccos \frac{1}{2} = \pi/3.$$



Definition:

$y = \arccos x \iff \cos y = x,$
$y \in [0, \pi]$

Eg. $\arccos(-1) = \pi$

$\arccos(\pi/2) = 0$

Cancellation properties

$\cos(\arccos x) = x$ if $x \in [-1, 1]$

$\arccos(\cos x) = x$ if $x \in [0, \pi]$

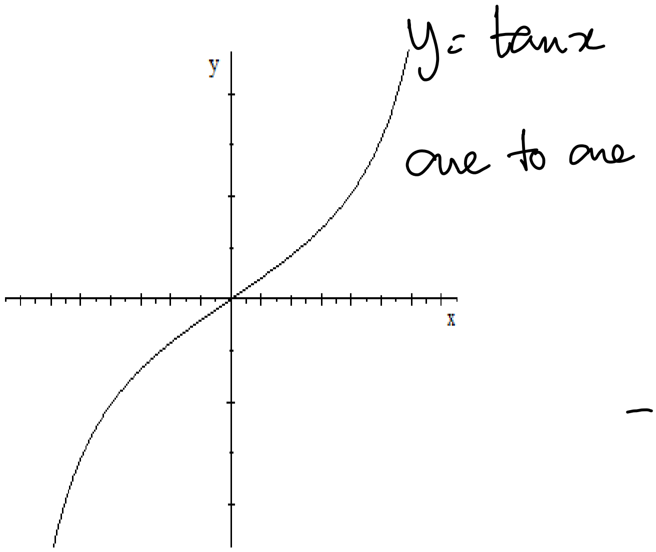
Eg.

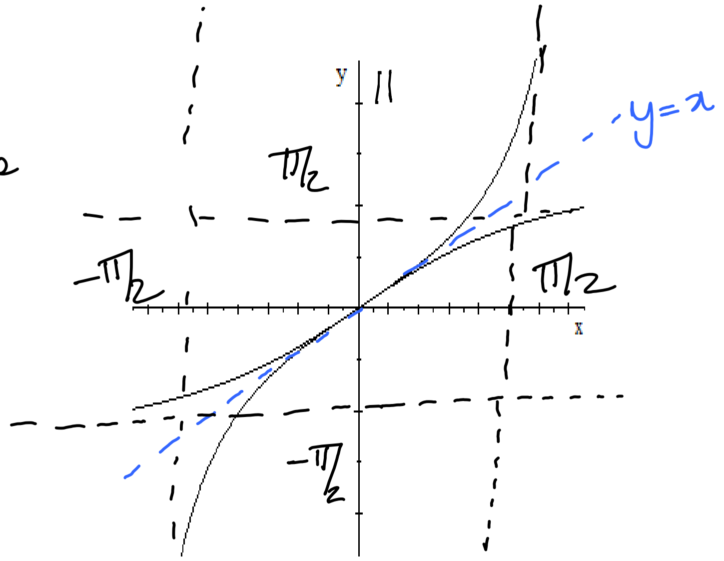
$\arccos(\cos \pi/4) = \pi/4$

$\cos^{-1}(\cos -\pi/4) = \frac{3\pi}{4}$

$\arccos\left(\cos \frac{-\pi}{6}\right) = \frac{5\pi}{6}$

$y = \arctan x$.





Definition

$$y = \arctan x \iff \tan y = x,\quad y \in \left(-\pi/2, \pi/2\right).$$

note the open brackets

Examples

1. $\arctan 1 = \pi/4$

2. $\arctan 0 = 0$

3. $\arctan -\sqrt{3} = -\dfrac{\pi}{3}$

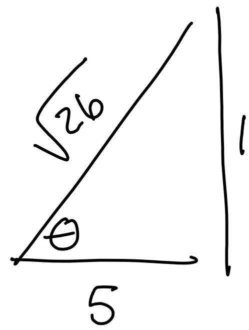
Cancellation properties:

$$\tan(\arctan x) = x, \quad x \in \mathbb{R}.$$

$$\arctan(\tan x) = x, \quad x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

More examples:

1. Find $\cos(\underbrace{\arctan \tfrac{1}{5}}_{\theta})$



$$= \frac{5}{\sqrt{26}}$$

2. Find $\arctan(\underbrace{\tan \tfrac{3\pi}{4}}_{-1})$

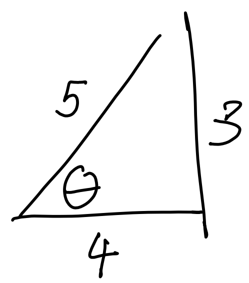
$$= -\frac{\pi}{4}$$

3.1 Solve $\tan x = -1$

$$x = \frac{3\pi}{4} + k\pi, \quad k \in \mathbb{Z}$$

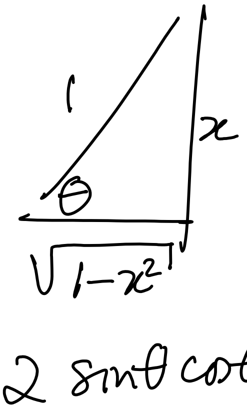
3.2 Find $\arctan(-1) = \frac{-\pi}{4}$

4. Find $\tan(\underbrace{\arcsin \frac{3}{5}}_{\theta})$



$$= \frac{3}{4}$$

5. $\sin(2\underbrace{\arcsin x}_{\theta})$



$$= \sin\theta\cos\theta + \cos\theta\sin\theta = 2\sin\theta\cos\theta$$

$$= 2\,\frac{x}{1} \times \frac{\sqrt{1-x^2}}{1} = 2x\sqrt{1-x^2}$$